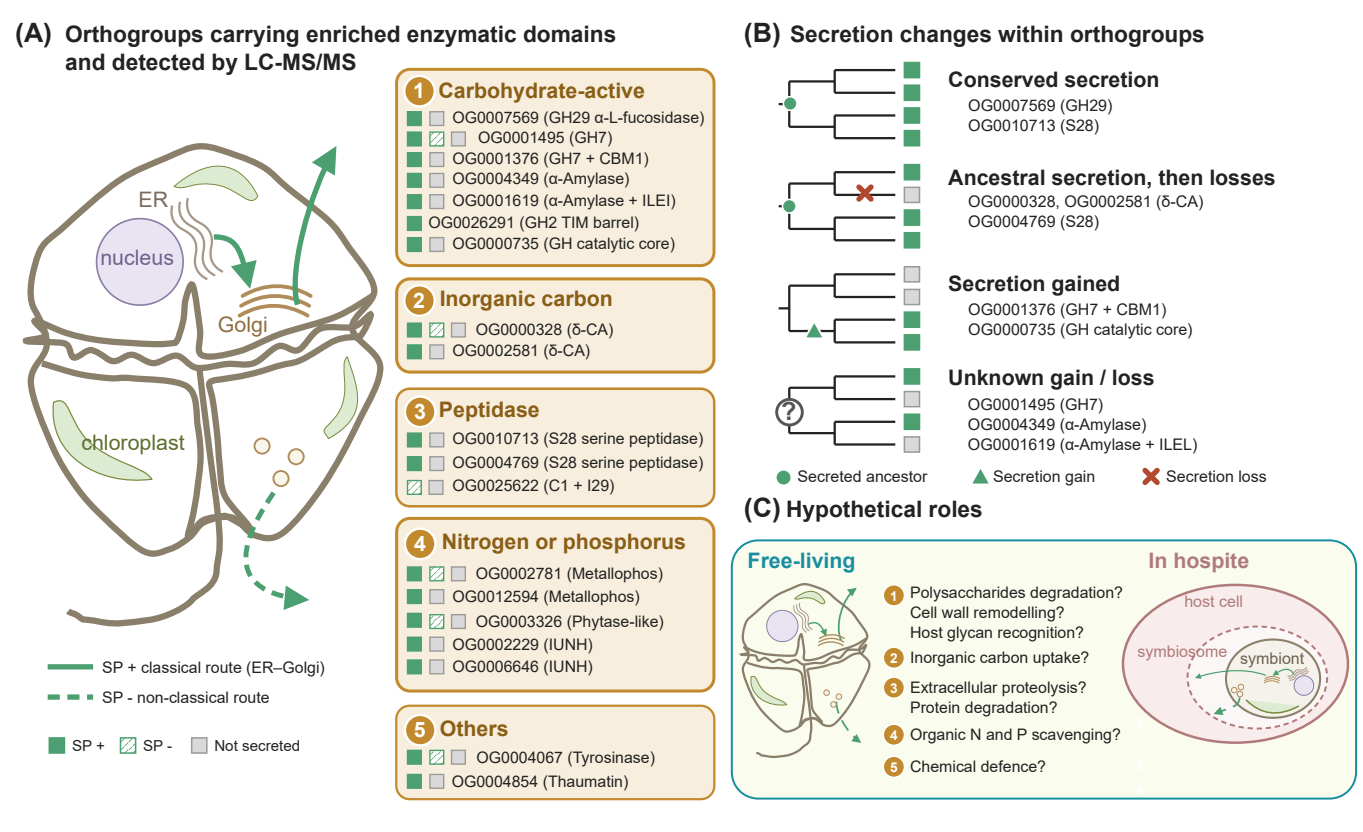

(A) Orthogroups carrying enriched enzymatic domains and detected by LC-MS/MS
ER
nucleus
Golgi
chloroplast
SP + classical route (ER–Golgi)
SP - non-classical route
SP +
SP -
Not secreted
1 Carbohydrate-active
OG0007569 (GH29 α-L-fucosidase)
OG0001495 (GH7)
OG0001376 (GH7 + CBM1)
OG0004349 (α-Amylase)
OG0001619 (α-Amylase + ILEI)
OG0026291 (GH2 TIM barrel)
OG0000735 (GH catalytic core)
2 Inorganic carbon
OG0000328 (δ-CA)
OG0002581 (δ-CA)
3 Peptidase
OG0010713 (S28 serine peptidase)
OG0004769 (S28 serine peptidase)
OG0025622 (C1 + I29)
4 Nitrogen or phosphorus
OG0002781 (Metallophos)
OG0012594 (Metallophos)
OG0003326 (Phytase-like)
OG0002229 (IUNH)
OG0006646 (IUNH)
5 Others
OG0004067 (Tyrosinase)
OG0004854 (Thaumatin)
(B) Secretion changes within orthogroups
Conserved secretion
OG0007569 (GH29)
OG0010713 (S28)
Ancestral secretion, then losses
OG0000328, OG0002581 (δ-CA)
OG0004769 (S28)
Secretion gained
OG0001376 (GH7 + CBM1)
OG0000735 (GH catalytic core)
Unknown gain / loss
OG0001495 (GH7)
OG0004349 (α-Amylase)
OG0001619 (α-Amylase + ILEL)
Secreted ancestor
Secretion gain
Secretion loss
(C) Hypothetical roles
Free-living
1 Polysaccharides degradation?
Cell wall remodelling?
Host glycan recognition?
2 Inorganic carbon uptake?
3 Extracellular proteolysis?
Protein degradation?
4 Organic N and P scavenging?
5 Chemical defence?
In hospite
host cell
symbiosome
symbiont